

Quantifying Moderator Effects on High-Dimensional Multi-Subject Time Series

Abstract

Researchers across many areas use time-series data to estimate relationships among variables over time. These relationships can be modeled via Granger-causal networks (GCNs), where nodes correspond to variables and directed edges indicate lagged temporal relationships among them. When time-series data are gathered over multiple subjects, GCN models must account for the presence of shared and subject-specific edges across the GCNs. While existing methods can estimate GCNs on multi-subject data, they often lack interpretable explanations for the source of the observed heterogeneity. In the present paper, we propose the moderator vector autoregressive model (MOD-VAR) for interpretable estimation of GCNs in multi-subject time-series data. To this end, we decompose the GCNs into shared, moderator-dependent, and subject-specific components. The use of moderator-dependent components can enhance both the quantification and interpretability of heterogeneous GCNs. Our theoretical results establish that MOD-VAR improves multi-subject GCN estimation in high-dimensional regimes with substantial heterogeneity. Furthermore, extensive numerical simulations demonstrate that MOD-VAR outperforms existing methods in terms of both GCN estimation and forecasting in a variety of settings. Applying MOD-VAR to a resting-state fMRI dataset, we recover moderating effects of age and locus coeruleus structural integrity on signaling within and between the salience, frontoparietal, and default mode networks.

Keywords: Adaptive Estimation, Granger-Causal Networks, Network Regression, Vector Autoregressive Models, Weighted Networks.

1 Introduction

The modeling and forecasting of high-dimensional time series data has been a long-standing area of research in statistics and machine learning [11], with applications in economics, finance, neuroscience, and psychology [14; 25; 5; 23; 15], among other areas. *Granger-causal networks* (GCNs) are a particularly useful way to describe relationships among variables in a time series. In a GCN, nodes correspond to variables, and a directed edge between two nodes represents a predictive effect of past values of the source variable on future values of the target variable. When the time dependence is linear, the GCN can be recovered from the support of the coefficients of a sparse vector autoregressive model (VAR) [3]. This has led to the development of a wide variety of methods aimed at accurately estimating GCNs in time series data [2; 3; 22]. For a review, see Wei [32]; Shojaie and Fox [26].

One major statistical challenge is that of estimating GCNs over multiple subjects. For example, in functional magnetic resonance imaging (fMRI) studies, researchers are often interested in determining the patterns of connectivity among different regions of the brain [14; 25; 5]. By processing fMRI data, researchers can measure the activity of brain regions over time. In this context, a GCN may describe directed temporal dependencies among the activity of different brain regions. In multi-subject studies, we expect the connectivity patterns to have shared components, while also showing some heterogeneity across different subjects. In this setting, fitting a single VAR model may lead to inaccurate GCN estimation and forecasting. On the other hand, fitting separate VAR models misses the opportunity to exploit the shared VAR structure across subjects, to improve estimation accuracy [6].

Several works have explored the problem of fitting and testing models for multi-subject high-dimensional time series. The Multi-VAR method [7; 8] applies penalized estimation to recover both shared and subject-specific components of the GCNs. The group iterative

multiple model estimation procedure (GIMME) creates subject-level networks with unified structural equation models for each subject and then identifies majority-shared network edges [9; 15; 16]. Recent works have also explored the task of testing the significance of sparse GCN estimators. For example, Zhu and Liu [36]; Uematsu and Yamagata [29] developed multiple testing procedures based on the debiased LASSO [30; 33; 13] to determine the significance of GCN connections for a single subject.

Despite the progress of prior works, a key question remains unanswered: *Can we explain the heterogeneity of GCNs across subjects?* For example, in the aforementioned fMRI application, researchers may be interested not only in detecting variability in the GCNs across individuals, but also in determining whether moderators like age, biological sex, or prior health conditions explain that variability. Existing methods that model multi-subject GCNs do not provide mechanisms for determining whether heterogeneity across GCNs can be explained by available external moderators.

Although models that link the GCNs of time-series data with moderator variables have been explored via multilevel modeling [27; 17], their applicability to high-dimensional settings is limited. In particular, they cannot accommodate a large number of variables or moderators without requiring a prohibitively large number of subjects or time measurements. Alternatively, the vector autoregressive model with exogenous variables (VARX) incorporates the effects of external moderators on the VAR model [22; 21]. Despite its advantages, external moderators are only considered as affecting the mean behavior of the time series, ignoring the potential effect of moderators on the GCNs themselves. Thus, there is a need for methods that reliably estimate how moderator variables influence GCNs across multiple subjects, especially in high-dimensional settings.

In the present work, we propose the moderator vector autoregressive model (MOD-

VAR) for the simultaneous estimation of GCNs on multi-subject time-series data with time-invariant moderators. MOD-VAR models how external moderators influence GCNs across subjects by decomposing the GCN of each subject into (i) edges shared across all subjects, (ii) edges dependent on external moderators, and (iii) subject-specific edges that cannot be explained by the moderator variables. Our method is highly general, encompassing the VAR and Multi-VAR frameworks [7; 8]. Although similar moderator models have been explored in the context of Gaussian graphical models [20; 34; 18], to the best of our knowledge this is the first method to decompose an autoregressive structure into shared, moderator-dependent, and subject-specific components to improve forecasting and interpretability of multi-subject high-dimensional time series.

The rest of the paper is organized as follows. In Section 2 we introduce our moderator time series model. In Section 3, we introduce our MOD-VAR estimator and its variants. We provide details on the implementation of our proposed method in Section 4. We explore theoretical properties of the MOD-VAR in Section 5. In Section 6 we provide numerical comparisons of our proposed MOD-VAR with existing methods found in the literature. In Section 7 we present the analysis of a dataset of brain activation across multiple regions via fMRI measurements, and quantify the effect of multiple moderators on brain connectivity across subjects. Closing remarks and discussions are found in Section 8.

Notational Conventions

We establish the following notational conventions. For $1 \leq k \leq \ell$, we denote $[k : \ell] := \{k, k+1, \dots, \ell-1, \ell\}$ and $[k] := [1 : k]$. For any set B , we denote the indicator function of B as $\mathbf{1}(B)$, and the cardinality of B as $|B|$. Given a matrix $A = (a_{ij})_{i \in [k], j \in [\ell]} \in \mathbb{R}^{k \times \ell}$, we denote the transpose of A as $A' \in \mathbb{R}^{\ell \times k}$. Let $B \subset [k]$, $C \subset [\ell]$, $i \in [k]$ and

$j \in [\ell]$. We denote $A_{B,\cdot} := (a_{ij})_{i \in B, j \in [\ell]} \in \mathbb{R}^{|B| \times \ell}$ and $A_{\cdot, C} := (a_{ij})_{i \in [k], j \in C} \in \mathbb{R}^{k \times |C|}$, $A_{i\cdot} := (a_{ij})_{j \in [\ell]} \in \mathbb{R}^\ell$ and $A_{\cdot j} := (a_{ij})_{i \in [k]} \in \mathbb{R}^k$. For $\mathbf{z} = (z_1, z_2, \dots, z_k)' \in \mathbb{R}^k$ we denote $\mathbf{z}_B := (z_b)_{b \in B} \in \mathbb{R}^{|B|}$. Furthermore, let $\|\mathbf{z}\|_1, \|\mathbf{z}\|_2, \|\mathbf{z}\|_\infty$ denote the L_1 , Euclidean and entrywise maximum norms respectively, given by $\|\mathbf{z}\|_1 := \sum_{i=1}^k |z_i|$, $\|\mathbf{z}\|_2 := \sqrt{\mathbf{z}'\mathbf{z}}$, and $\|\mathbf{z}\|_\infty := \max_{i \in [k]} |z_i|$. Given $A = (a_{ij})_{i,j \in [k]} \in \mathbb{R}^{k \times k}$, we denote the trace as $\text{tr}(A) := \sum_{i \in [k]} a_{ii}$. If A is symmetric, we denote the minimum and maximum eigenvalues of A as $\Lambda_{\min}(A), \Lambda_{\max}(A) \in \mathbb{R}$, respectively. For $A \in \mathbb{R}^{k \times \ell}$, we denote by $\|A\|_2, \|A\|_F$ and $\|A\|_1$ the L_2 operator, Frobenius and vectorized L_1 norms, respectively, given by $\|A\|_2 := \sup_{\mathbf{z} \in \mathbb{R}^\ell \setminus \{\mathbf{0}\}} \|A\mathbf{z}\|_2 / \|\mathbf{z}\|_2$, $\|A\|_F := \sqrt{\text{tr}(A'A)}$ and $\|A\|_1 := \sum_{i,j} |a_{ij}|$. Furthermore, for $B \in \mathbb{R}^{m \times g}$, we define the matrix Kronecker product $A \otimes B \in \mathbb{R}^{km \times \ell g}$ via the expression $A \otimes B = (a_{ij}B)_{i \in [k], j \in [\ell]}$. For any k , we denote $\mathbf{1}_k \in \mathbb{R}^k$ the vector with all entries equal to one. For any $z \in \mathbb{R}$, let $\text{sign}(z) := \mathbf{1}(z > 0) - \mathbf{1}(z < 0)$.

2 Moderator Vector Autoregressive Model

2.1 Vector Autoregressive Models

Consider a study with n subjects. For each subject $k \in [n]$, we observe d different variables of interest over T_k equally spaced time points $\mathcal{Y}_k = \{\mathbf{Y}_t^{(k)}\}_{t=1}^{T_k}$, where $\mathbf{Y}_t^{(k)} = (Y_{t1}^{(k)}, Y_{t2}^{(k)}, \dots, Y_{td}^{(k)})' \in \mathbb{R}^d$. We model the lagged temporal relationships that exist between the variables by a VAR(1) model, expressed as

$$\mathbf{Y}_t^{(k)} = \Phi^{(k)*} \mathbf{Y}_{t-1}^{(k)} + \boldsymbol{\varepsilon}_t^{(k)}, \quad \text{for all } 2 \leq t \leq T_k,$$

where the matrix $\Phi^{(k)*} \in \mathbb{R}^{d \times d}$ encodes the true lag-1 multivariate relationships for subject k , and the error terms follow a multivariate Gaussian distribution, *i.e.* $\{\boldsymbol{\varepsilon}_t^{(k)} \in \mathbb{R}^d : 1 < t \leq T_k\} \stackrel{i.i.d.}{\sim} N_d(\mathbf{0}, \Sigma_\varepsilon^{(k)})$, where the matrices $\Sigma_\varepsilon^{(1)}, \Sigma_\varepsilon^{(2)}, \dots, \Sigma_\varepsilon^{(n)} \in \mathbb{R}^{d \times d}$ are positive definite.

In particular, for $1 \leq i, j \leq d$, the entry $\Phi_{ij}^{(k)*}$ characterizes the predictive effect of the variable $Y_j^{(k)}$ at time $(t-1)$ on the variable $Y_i^{(k)}$ at time t . Observe that we define a distinct VAR coefficient matrix $\Phi^{(k)*}$ for each subject $k \in [n]$. Given the matrix $\Phi^{(k)*}$, we define the GCN associated with subject k as a network $\mathcal{G}^{(k)} = (\mathcal{V}, \mathcal{E}^{(k)})$, with a set of nodes $\mathcal{V} = [d]$, and a directed edge $i \rightarrow j$ is present in $\mathcal{E}^{(k)}$ if and only if $\Phi_{ji}^{(k)*} \neq 0$. Thus, the structure of the GCN $\mathcal{G}^{(k)}$ depends on the sparsity pattern of the matrix $\Phi^{(k)*}$ [29].

For each subject $k = 1, 2, \dots, n$, in addition to the time series data $\mathcal{Y}_k$, we have a set of p potential moderator variables $\mathbf{X}_k = (X_{k1}, X_{k2}, \dots, X_{kp})' \in \mathbb{R}^p$. The moderators in $\mathbf{X}_k$ are time-invariant, and thus can be thought of as stable features of each subject. For example, in a neuroimaging study analyzing brain connectivity, these moderators may include demographic variables such as biological sex or age. We may also consider controlled experimental variables of interest such as treatment status of an individual, or dose magnitude provided to a patient. Our goal is to quantify the effect that these time-invariant moderator variables have on the GCN connectivity patterns across subjects to enhance model interpretation and forecasting accuracy.

2.2 Moderator Vector Autoregressive Models

We first consider the setting in which all heterogeneity across subjects can be explained by the moderators in $\mathbf{X}_k$. In this case, we model the VAR coefficients of subject k as $\Phi^{(k)*} = \Phi^*(\mathbf{X}_k)$, where, for any vector of p moderators $\mathbf{x} = (x_1, x_2, \dots, x_p)' \in \mathbb{R}^p$,

$$\Phi^*(\mathbf{x}) := \Phi^{(0)*} + x_1 \Phi^{(M1)*} + x_2 \Phi^{(M2)*} + \dots + x_p \Phi^{(Mp)*}, \quad (1)$$

where $\Phi^{(0)*}, \Phi^{(M1)*}, \Phi^{(M2)*}, \dots, \Phi^{(Mp)*} \in \mathbb{R}^{d \times d}$. The matrix $\Phi^{(0)*} \in \mathbb{R}^{d \times d}$ corresponds to autoregressive effects shared across all subjects. For each moderator variable $1 \leq r \leq p$, the matrix $\Phi^{(Mr)*}$ characterizes the effect that the r -th moderator has on the time-series

dynamics. More specifically, for $i, j \in [d]$ and $r \in [p]$ the entry $\Phi_{ij}^{(Mr)*}$ represents the effect that the r -th moderator variable has on the lagged relationship of the variable Y_j at time $(t-1)$ on the variable Y_i at time t . For example, if a subject has moderator vector $\mathbf{x} = (x_1, x_2, \dots, x_p)' \in \mathbb{R}^p$ with $x_r = 0$, we have that $x_r \Phi^{(Mr)*} = \mathbf{0}$. Thus, the magnitude of autoregressive effects related to moderator r for this subject is zero. In general, if $\Phi^{(Mr)*} \neq 0$, the magnitude of x_r modulates the magnitude of the term $x_r \Phi^{(Mr)*}$ in $\Phi^*(\mathbf{x})$. Under the model (1), we define the moderator vector autoregressive (MOD-VAR) model via the equation

$$\mathbf{Y}_t^{(k)} = \Phi^*(\mathbf{X}_k) \mathbf{Y}_{t-1}^{(k)} + \boldsymbol{\varepsilon}_t^{(k)}, \quad \text{for all } 1 \leq k \leq n \text{ and } 2 \leq t \leq T_k. \quad (2)$$

We illustrate our MOD-VAR model from (2) with the following example.

Example 1 (Binary moderators). *We consider a MOD-VAR model with n subjects and a single moderator variable (i.e., $p = 1$) $X_1, X_2, \dots, X_n \in \mathbb{R}$ with $\{X_k\}_k \stackrel{i.i.d.}{\sim} \text{Bernoulli}(w)$, for some $0 < w < 1$. Our proposed model can be written as,*

$$\mathbf{Y}_t^{(k)} = [\Phi^{(0)*} + X_k \Phi^{(M1)*}] \cdot \mathbf{Y}_{t-1}^{(k)} + \boldsymbol{\varepsilon}_t^{(k)}, \quad \text{for all } 2 \leq t \leq T_k, \text{ and } k \in [n].$$

The VAR coefficient matrices of all subjects share the term $\Phi^{(0)} \in \mathbb{R}^{d \times d}$. Furthermore, the presence of the matrix $\Phi^{(M1)*}$ in the k -th subject model depends on the value of X_k . All subjects with $X_k = 1$ have their GCN associated with the support of $\Phi^{(k)*} = \Phi^{(0)*} + \Phi^{(M1)*}$, while subjects with $X_k = 0$ have their GCN associated with the support of $\Phi^{(k)*} = \Phi^{(0)*}$. Thus, the matrix $\Phi^{(M1)*}$ captures the structural differences of the time-series dynamics across the grouping given by the moderators $X_1, X_2, \dots, X_n \in \{0, 1\}$.*

Now, consider modeling two binary moderator variables $\mathbf{X}_k = (X_{k1}, X_{k2})' \in \{0, 1\}^2$. We now have a total of four possible GCNs, associated with the support of the following matrices: (i) $\Phi^{(k)} = \Phi^{(0)*}$ if $\mathbf{X}_k = (0, 0)'$, (ii) $\Phi^{(k)*} = \Phi^{(0)*} + \Phi^{(M1)*}$ if $\mathbf{X}_k = (1, 0)'$, (iii) $\Phi^{(k)*} = \Phi^{(0)*} + \Phi^{(M2)*}$ if $\mathbf{X}_k = (0, 1)'$, and (iv) $\Phi^{(k)*} = \Phi^{(0)*} + \Phi^{(M1)*} + \Phi^{(M2)*}$ if*

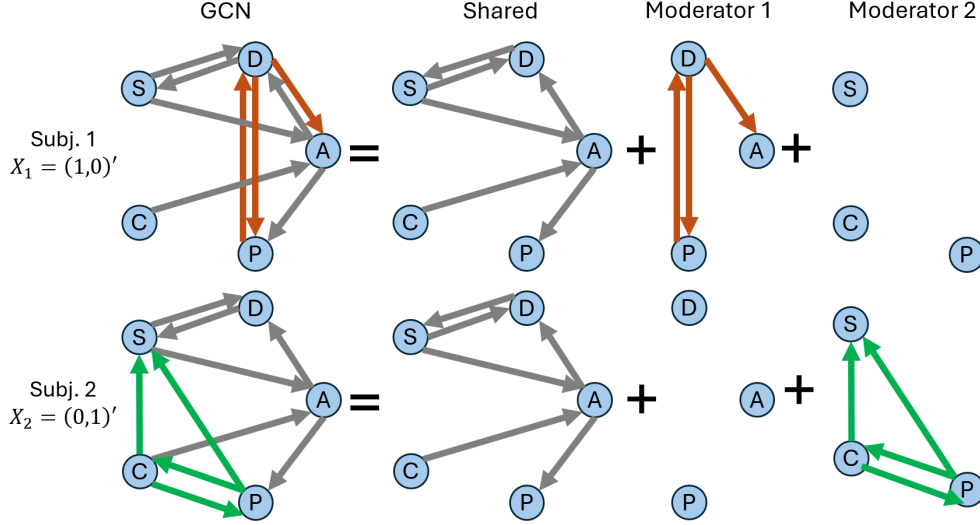


Figure 1: Illustration of the GCN of two subjects according to the MOD-VAR framework with $d = 5$ variables and $p = 2$ binary moderators, as discussed in Example 1. Top-panels: visualization of the decomposition of the GCN of subject 1. Since $\mathbf{X}_1 = (1, 0)'$, only the first moderator contributes a non-zero term to the GCN of this subject. Bottom panels: visualization of the GCN for subject 2. Since $\mathbf{X}_2 = (0, 1)'$, only the second moderator contributes a non-zero term to this subject's GCN.

$\mathbf{X}_k = (1, 1)'$. In Figure 1, we visualize two GCNs associated with different subjects, with $d = 5$ variables and $p = 2$ binary moderators.

2.3 MOD-VAR with Subject-Specific Components

Although external moderators may account for some of the GCN heterogeneity across subjects, important subject-specific edges may remain unexplained by the observed moderators. The model introduced in (2) does not account for the presence of such subject-specific edges. In real-data applications where it is of interest to understand subject-specific dynamics, not accounting for such systematic effects may lead to incorrect interpretations of the results and higher forecasting errors. Thus, in addition to a shared component $\Phi^{(0)*}$, and moderator-dependent components $\Phi^{(M1)*}, \dots, \Phi^{(Mp)*}$, we introduce the matrices $\Delta^{(I1)*}, \Delta^{(I2)*}, \dots, \Delta^{(In)*} \in \mathbb{R}^{d \times d}$ to model subject-specific autoregressive effects. For each

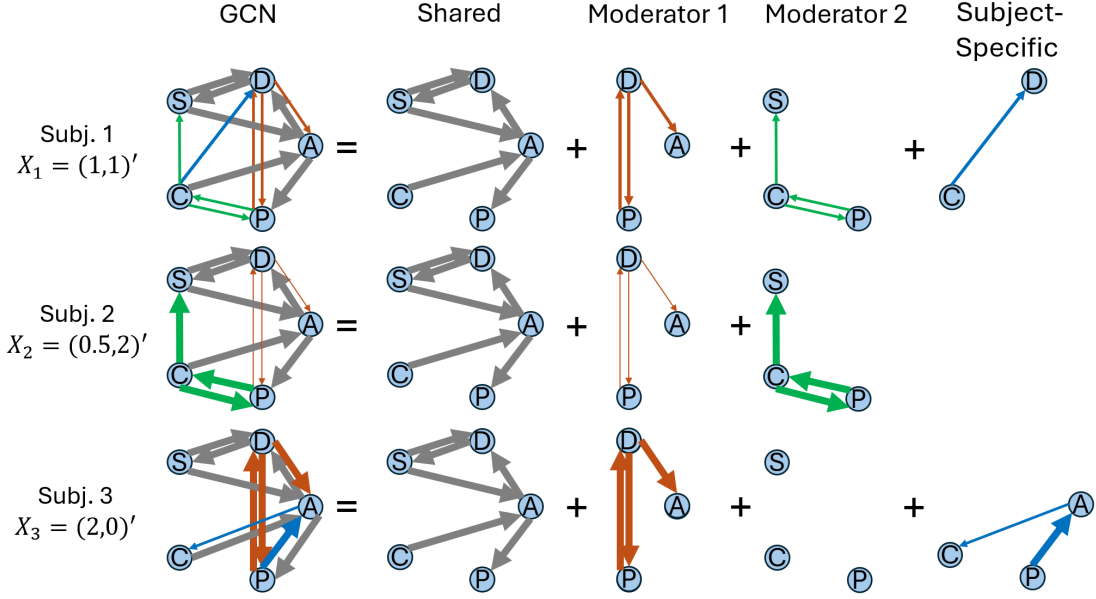


Figure 2: Illustration of the MOD-VAR framework with $d = 5$ variables, measured on $n = 3$ subjects, with $p = 2$ continuous moderators and subject-specific terms $\{\Delta^{(Ik)*}\}_{k \in [n]}$. Some GCN edges are present for all subjects. Some edges have their strength associated with the moderator value of the corresponding subject. Additional subject-specific edges may be present in some GCNs.

subject k , the coefficient matrix $\Phi^{(k)*}$ can be decomposed as

$$\Phi^{(k)*} = \Phi^*(\mathbf{X}_k) + \Delta^{(Ik)*} = \Phi^{(0)*} + \left(\sum_{r=1}^p X_{kr} \Phi^{(Mr)*} \right) + \Delta^{(Ik)*}. \quad (3)$$

From this, the MOD-VAR model with subject-specific components is given by,

$$\mathbf{Y}_t^{(k)} = (\Phi^*(\mathbf{X}_k) + \Delta^{(Ik)*}) \mathbf{Y}_{t-1}^{(k)} + \boldsymbol{\varepsilon}_t^{(k)}, \quad \text{for all } 1 \leq k \leq n \text{ and } 2 \leq t \leq T_k. \quad (4)$$

The term $\Delta^{(Ik)*} \in \mathbb{R}^{d \times d}$ only appears in the autoregressive equation of subject k . Thus, the matrix $\Delta^{(Ik)*}$ encodes the autoregressive effects of subject k not captured by the shared or moderator-dependent components.

We visualize model (4) for $n = 3$ subjects, $d = 5$ variables, $p = 2$ moderators, and subject-specific components in Figure 2. Our goal is to estimate the shared, moderator-dependent, and subject-specific components $\Phi^{(0)*}$, $\{\Phi^{(Mr)*}\}_{r=1}^p$ and $\{\Delta^{(Ik)*}\}_{k=1}^n$, respectively.

3 Methodology

We introduce our method for simultaneously estimating VAR coefficient matrices associated with the shared, moderator-dependent and subject-specific components on multi-subject time series data, given by $\Phi^{(0)*}$, $\{\Phi^{(Mr)*}\}_{r=1}^p$ and $\{\Delta^{(Ik)*}\}_{k=1}^n$, respectively. We achieve this via an L_1 -penalized estimation approach. We first introduce the method under the assumption of no subject-specific components, *i.e.*, $\Delta^{(I1)*} = \Delta^{(I2)*} = \dots = \Delta^{(In)*} = \mathbf{0}$.

3.1 MOD-VAR With Only Moderator Components

Consider the case in which all GCN heterogeneity is induced by moderator-dependent effects, as described in (1). Then, our MOD-VAR model follows the structure (2). Let $\Phi = [\Phi^{(0)}, \Phi^{(M1)}, \Phi^{(M2)}, \dots, \Phi^{(Mp)}] \in \mathbb{R}^{d \times (d+pd)}$, and, for any vector of p moderators $\mathbf{x}$, we define $\Phi(\mathbf{x})$ as the $d \times d$ matrix given by

$$\Phi(\mathbf{x}) := \Phi^{(0)} + x_1 \Phi^{(M1)} + x_2 \Phi^{(M2)} + \dots + x_p \Phi^{(Mp)}. \quad (5)$$

We use Φ to denote both the matrix and its induced operator $\Phi(\mathbf{x})$ from (5).

Our goal is to estimate the matrix Φ^* , consisting of a total of $(p+1)d^2$ parameters. This estimation may be challenging, especially in cases where the number of subjects n or number of time points per subject $\{T_k\}_{k=1}^n$ are limited. In such scenarios, one may estimate Φ^* under the assumption of sparsity, *i.e.*, only a small number of parameters in Φ^* are non-zero [2; 7]. Under this assumption, we derive a sparse estimation of Φ^* via the optimization problem

$$\hat{\Phi} = \underset{\Phi \in \mathbb{R}^{d \times (d+pd)}}{\operatorname{argmin}} \frac{1}{2N} \sum_{k=1}^n \sum_{t=2}^{T_k} \|\mathbf{Y}_t^{(k)} - \Phi(\mathbf{X}_k) \mathbf{Y}_{t-1}^{(k)}\|_2^2 + \lambda_1 \|\Phi^{(0)}\|_1 + \lambda_2 \sum_{r=1}^p \|\Phi^{(Mr)}\|_1. \quad (6)$$

where $N = T_1 + T_2 + \dots + T_n - n$, and $\lambda_1, \lambda_2 > 0$ are tuning parameters. The optimization (6) aims to minimize the in-sample mean square forecasting error (MSFE) across all subjects.

Furthermore, it incorporates a non-uniform penalty that induces different levels of sparsity for the shared component $\Phi^{(0)}$ and the moderator-dependent components $\{\Phi^{(Mr)}\}_{r \in [p]}$.

The optimization problem (6) can be written in a more compact matrix form. For this, consider $\mathbf{Z}^{(k)} = [\mathbf{Y}_{T_k}^{(k)}, \mathbf{Y}_{T_k-1}^{(k)}, \dots, \mathbf{Y}_2^{(k)}]'$, $\mathbf{Y}^{(k)} = [\mathbf{Y}_{T_k-1}^{(k)}, \mathbf{Y}_{T_k-2}^{(k)}, \dots, \mathbf{Y}_1^{(k)}]'$, and $\mathbf{E}^{(k)} = [\boldsymbol{\varepsilon}_{T_k}^{(k)}, \boldsymbol{\varepsilon}_{T_k-1}^{(k)}, \dots, \boldsymbol{\varepsilon}_2^{(k)}]'$. Here the matrices $\mathbf{Z}^{(k)}, \mathbf{Y}^{(k)}, \mathbf{E}^{(k)} \in \mathbb{R}^{(T_k-1) \times d}$ aggregate the contemporaneous, lagged, and error terms for subject k , respectively. Furthermore, we introduce

$$\mathbf{W}^{(k)} := [\mathbf{Y}^{(k)}, X_{k1} \mathbf{Y}^{(k)}, \dots, X_{kp} \mathbf{Y}^{(k)}] = (\mathbf{X}_k^*)' \otimes \mathbf{Y}^{(k)} \in \mathbb{R}^{(T_k-1) \times (d+pd)},$$

where $\mathbf{X}_k^* = (1, \mathbf{X}_k')' \in \mathbb{R}^{p+1}$. One can show that $\mathbf{Y}^{(k)} \cdot (\Phi^*(\mathbf{X}_k))' = (\mathbf{W}^{(k)}) \cdot (\Phi^*)'$. We gather data from all subjects through the matrices $\mathbf{Z} := [\mathbf{Z}^{(1)'}, \mathbf{Z}^{(2)'}, \dots, \mathbf{Z}^{(n)'}]'$, $\mathbf{W} := [\mathbf{W}^{(1)'}, \mathbf{W}^{(2)'}, \dots, \mathbf{W}^{(n)'}]'$, and $\mathbf{E} := [\mathbf{E}^{(1)'}, \mathbf{E}^{(2)'}, \dots, \mathbf{E}^{(n)'}]'$. With this notation, we summarize the relationships in (2) with the equation $\mathbf{Z} = \mathbf{W} \cdot (\Phi^*)' + \mathbf{E}$. Thus, the optimization problem (6) is equivalent to solving

$$\hat{\Phi} := \underset{\Phi = [\Phi^{(0)}, \Phi^{(M1)}, \dots, \Phi^{(Mp)}]}{\operatorname{argmin}} \frac{1}{2N} \|\mathbf{Z} - \mathbf{W}\Phi'\|_F^2 + \lambda_1 \|\Phi^{(0)}\|_1 + \lambda_2 \|\Phi^{(M1)}, \dots, \Phi^{(Mp)}\|_1. \quad (7)$$

The estimation of $\hat{\Phi}$ can be reduced to a multiple regression optimization problem, with a non-uniform L_1 penalty. We describe the optimization algorithm in Section 4.

3.2 MOD-VAR With Subject-Specific Components

In many real-data scenarios, subject-specific edges in the GCN not associated with the available moderator data may be present for some subjects. In such cases, one may model the presence of subject-specific components in the GCNs. We construct a set of artificial moderators to adapt the optimization problem described in Section 3.1 for estimating common, moderator-dependent and subject-specific components of the GCNs.

First, we consider the case in which we do not have moderator data available, and simply aim to estimate the shared and subject-specific VAR coefficients across multiple

subjects. This scenario corresponds to the problem studied by the Multi-VAR method [7]. For each subject k , we define the artificial moderator vector $\mathbf{X}_k^{(I)} = \mathbf{1}_k \in \mathbb{R}^n$, a column vector of size $n \times 1$, with zero in all entries except the k -th entry. Since $\mathbf{X}_k^{(I)}$ has n artificial moderators, the associated parameters of the MOD-VAR model can be expressed in the form of a $d \times (d + nd)$ matrix $\Phi = [\Phi^{(0)}, \Phi^{(M1)}, \Phi^{(M2)}, \dots, \Phi^{(Mn)}]$. Since $\mathbf{X}_k^{(I)}$ is an indicator vector for the k -th entry and (5), $\Phi(\mathbf{X}_k^{(I)}) = \Phi^{(0)} + \Phi^{(Mk)}$, for all $k \in [n]$. We note that the term $\Phi^{(Mk)}$ appears only in the coefficients of subject k . Therefore, $\Phi^{(Mk)}$ can be interpreted to represent autoregressive effects specific to subject k . From this, we change our notation for the estimating parameter to $\Phi = [\Phi^{(0)}, \Delta^{(I1)}, \Delta^{(I2)}, \dots, \Delta^{(In)}]$, where $\Delta^{(Ik)}$ captures the autoregressive effects specific to subject k . In summary, the estimation of $\Phi = [\Phi^{(0)}, \Delta^{(I1)}, \Delta^{(I2)}, \dots, \Delta^{(In)}]$ can be achieved by solving our proposed MOD-VAR optimization problem (7) with the artificial moderator design matrix $\mathbf{X}_{(I)} := [\mathbf{X}_1^{(I)}, \mathbf{X}_2^{(I)}, \dots, \mathbf{X}_n^{(I)}]' = \mathbf{I}_n \in \mathbb{R}^{n \times n}$.

We also consider scenarios where the GCNs incorporate shared, moderator-dependent and subject-specific components, as described in Section 2.3. For this, we denote by $\mathbf{X}_{(M)} = [\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n]' \in \mathbb{R}^{n \times p}$ the moderator data matrix, and let $\mathbf{X}_{(I)} \in \mathbb{R}^{n \times n}$ be the artificial moderator matrix. We construct the expanded matrix $\mathbf{X}_{(M,I)} := [\mathbf{X}_{(M)}, \mathbf{X}_{(I)}]$. By augmenting the moderator matrix with artificial moderators we can simultaneously estimate the shared, moderator-dependent and subject-specific components through our proposed optimization problem (7).

4 Computational Algorithm and Estimation

We now discuss the procedure for optimizing (7), and selecting the optimal tuning parameters λ_1 and λ_2 for the MOD-VAR method.

4.1 Column-wise FISTA

The estimation of MOD-VAR requires solving the optimization problem (7). Since both terms $\|\mathbf{Z} - \mathbf{W}\Phi'\|_F^2/2N$ and $\mathcal{P}(\Phi) = \lambda_1 \|\Phi^{(0)}\|_1 + \lambda_2 \|[\Phi^{(M1)}, \dots, \Phi^{(Mp)}]\|_1$ admit an additive decomposition with respect to the rows of Φ , we can estimate $\hat{\Phi}$ by optimizing each row separately. Thus, for each $\ell \in [d]$ we aim to solve

$$\hat{\phi}_\ell = \underset{\phi \in \mathbb{R}^{d+pd}}{\operatorname{argmin}} \left\{ \frac{1}{2N} \|\mathbf{Z}_{\cdot\ell} - \mathbf{W}\phi\|_2^2 + \lambda_1 \|\phi_{[1:d]}\|_1 + \lambda_2 \|\phi_{[(d+1):(d+pd)]}\|_1 \right\}. \quad (8)$$

Once we obtain $\hat{\phi}_1, \hat{\phi}_2, \dots, \hat{\phi}_d \in \mathbb{R}^{d+pd}$, we construct $\hat{\Phi} = [\hat{\phi}_1, \hat{\phi}_2, \dots, \hat{\phi}_d]' \in \mathbb{R}^{d \times (d+pd)}$.

Then, the estimated shared component of the GCNs is defined as $\hat{\Phi}^{(0)} := \hat{\Phi}_{\cdot, [1:d]}$, and, for $r \in [p]$, the estimated component associated with the r -th moderator is defined as $\hat{\Phi}^{(Mr)} := \hat{\Phi}_{\cdot, [(rd+1):(d+rd)]}$. A summary of this procedure is provided in Algorithm 1. As discussed in Section 3.2, subject-specific components can be incorporated into the model via artificial moderator variables. Therefore, the procedure described in Algorithm 1 can be applied for estimating shared, moderator-dependent, and subject-specific components across GCNs, by simple modifications of the matrices $\mathbf{X}$ and $\mathbf{W}$.

In order to solve the optimization problem (8) for each $\ell \in [d]$, we apply the fast iterative shrinkage-thresholding algorithm (FISTA) [4]. For each $\ell \in [d]$, we define $\mathcal{L}^{(\ell)}(\phi) = \phi' \mathbf{W}' \mathbf{W} \phi / 2N - \mathbf{Z}'_{\cdot\ell} \mathbf{W} \phi / N$, and the penalty term $\mathcal{P}(\phi) = \lambda_1 \|\phi_{[1:d]}\|_1 + \lambda_2 \|\phi_{[(d+1):(d+pd)]}\|_1$. It can be shown that optimizing (8) is equivalent to solving $\operatorname{argmin}_\phi \mathcal{L}^{(\ell)}(\phi) + \mathcal{P}(\phi)$. Furthermore, we select a Lipschitz constant $L > 0$ for the gradient $\nabla \mathcal{L}^{(\ell)}(\phi) = (\mathbf{W}' \mathbf{W} / N) \phi - \mathbf{W}' \mathbf{Z}_{\cdot\ell} / N$. The inputs of our FISTA optimization procedure are $\mathbf{M} := \mathbf{W}' \mathbf{W} / N$, $\mathbf{b}_\ell := \mathbf{W}' \mathbf{Z}_{\cdot\ell} / N$, the tuning parameters $\lambda_1, \lambda_2 > 0$, the Lipschitz constant $L := \|\mathbf{W}' \mathbf{W}\|_2 / N$, a maximum iteration number I , and a tolerance level $\varepsilon > 0$. In addition, we may improve the efficiency of our estimation by utilizing a warm start $\phi_0 \in \mathbb{R}^{d+pd}$.

Algorithm 1: MOD-VAR.

Input: Time series $\{\mathbf{Y}_t^{(k)} : k \in [n], t \in [T_k]\} \subset \mathbb{R}^d$, moderators $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n]' \in \mathbb{R}^{n \times p}$, tuning parameters $\lambda_1, \lambda_2 > 0$.
for $k \in [n]$ **do**
 Define $\mathbf{Z}^{(k)} = [\mathbf{Y}_2^{(k)}, \mathbf{Y}_3^{(k)}, \dots, \mathbf{Y}_{T_k}^{(k)}]' \in \mathbb{R}^{(T_k-1) \times d}$.
 Define $\mathbf{Y}^{(k)} = [\mathbf{Y}_1^{(k)}, \mathbf{Y}_2^{(k)}, \dots, \mathbf{Y}_{T_k-1}^{(k)}]' \in \mathbb{R}^{(T_k-1) \times d}$.
 Set $\mathbf{X}_k^* := (1, \mathbf{X}'_k)' \in \mathbb{R}^{p+1}$.
 Define $\mathbf{W}^{(k)} = (\mathbf{X}_k^*)' \otimes \mathbf{Y}^{(k)} = [\mathbf{Y}^{(k)}, X_{k1}\mathbf{Y}^{(k)}, \dots, X_{kp}\mathbf{Y}^{(k)}] \in \mathbb{R}^{(T_k-1) \times (d+pd)}$.
end
Define $\mathbf{Z} = [\mathbf{Z}^{(1)'}, \mathbf{Z}^{(2)'}, \dots, \mathbf{Z}^{(n)'}]' \in \mathbb{R}^{N \times d}$.
Define $\mathbf{W} = [\mathbf{W}^{(1)'}, \mathbf{W}^{(2)'}, \dots, \mathbf{W}^{(n)'}]' \in \mathbb{R}^{N \times (d+pd)}$.
for $\ell \in [d]$ **do**
 Solve $\hat{\phi}_\ell = \operatorname{argmin}_{\phi \in \mathbb{R}^{d+pd}} \|\mathbf{Z}_{\cdot\ell} - \mathbf{W}\phi\|_2^2/2N + \lambda_1\|\phi_{[1:d]}\|_1 + \lambda_2\|\phi_{[(d+1):(d+pd)]}\|_1$.
end
Define $\hat{\Phi} := \hat{\Phi}(\lambda_1, \lambda_2) = [\hat{\phi}_1, \hat{\phi}_2, \dots, \hat{\phi}_d]' \in \mathbb{R}^{d \times (d+pd)}$.
Set $\hat{\Phi}^{(0)} := \hat{\Phi}^{(0)}(\lambda_1, \lambda_2) = \hat{\Phi}_{\cdot, [1:d]}$.
For $r \in [p]$, define $\hat{\Phi}^{(Mr)} := \hat{\Phi}^{(Mr)}(\lambda_1, \lambda_2) = \hat{\Phi}_{\cdot, [(rd+1):(d+rd)]}$.
Output: Common effects $\hat{\Phi}^{(0)}$, and moderated effects $\{\hat{\Phi}^{(Mr)}\}_{r=1}^p$

Our algorithm consists of the following steps. We start by setting $s_0 = 1$, and initializing $\phi_0 = \mathbf{0}$ if a warm start is not provided. Then, we set $\mathbf{y}_0 = \phi_0$, and define the vector of penalization weights as $\mathbf{w} = (\lambda_1 \mathbf{1}'_d, \lambda_2 \mathbf{1}'_{pd})' \in \mathbb{R}^{d+pd}$. Then, for any $i = 1, 2, \dots, I$, the updates for ϕ_i , s_i and $\mathbf{y}_i$ are given by the equations

$$\begin{aligned}\phi_i &:= ST_{\mathbf{w}/L}(\mathbf{y}_{i-1} - \nabla \mathcal{L}^{(\ell)}(\mathbf{y}_{i-1})/L), \\ s_i &:= (1 + \sqrt{1 + 4s_{i-1}^2})/2, \\ \mathbf{y}_i &:= \phi_{i-1} + ((s_{i-1} - 1)/s_i)(\phi_i - \phi_{i-1}),\end{aligned}$$

where, for any $\mathbf{w}, \mathbf{a} \in \mathbb{R}^m$, we define $ST_{\mathbf{w}}(\mathbf{a}) \in \mathbb{R}^m$ as the weighted entrywise soft thresholding function given by the entrywise value $[ST_{\mathbf{w}}(\mathbf{a})]_i = \operatorname{sign}(a_i) \cdot \max\{0, |a_i| - |w_i|\}$. We stop if $\|\phi_i - \phi_{i-1}\|_2 < \varepsilon$, or if we reach the maximum number of iterations. Then, we define $\hat{\phi}_\ell = \phi_i$. The full procedure is summarized in Algorithm 2.

Algorithm 2: FISTA for ℓ -step of MOD-VAR estimation.

Input: $\mathbf{M} := \mathbf{W}'\mathbf{W}/N \in \mathbb{R}^{(d+pd) \times (d+pd)}$, $\mathbf{b}_\ell := \mathbf{W}'\mathbf{Z}_\ell/N \in \mathbb{R}^{d+pd}$, hyperparameters $\lambda_1, \lambda_2 > 0$, Lipschitz constant $L := \|\mathbf{M}\|_2$, maximum iterations $I > 0$, tolerance $\varepsilon > 0$, warm start $\boldsymbol{\phi}_0 \in \mathbb{R}^{d+pd}$.

Initialize $\boldsymbol{\phi}_0 = \mathbf{0} \in \mathbb{R}^{d+pd}$ if warm start not provided.

Initialize $s_0 = 1$, $\mathbf{y}_0 = \boldsymbol{\phi}_0$, $\mathbf{w} = (\lambda_1 \mathbf{1}'_d, \lambda_2 \mathbf{1}'_{pd})' \in \mathbb{R}^{d+pd}$ and $i = 0$.

do

- Set $i \leftarrow i + 1$.
- Calculate $\nabla \mathcal{L}^{(\ell)}(\mathbf{y}_{i-1}) = \mathbf{M}\mathbf{y}_{i-1} - \mathbf{b}_\ell$.
- Set $\boldsymbol{\phi}_i := ST_{\mathbf{w}/L}(\mathbf{y}_{i-1} - \nabla \mathcal{L}^{(\ell)}(\mathbf{y}_{i-1})/L)$.
- Set $s_i = (1 + \sqrt{1 + 4s_{i-1}^2})/2$.
- Set $\mathbf{y}_i = \boldsymbol{\phi}_{i-1} + ((s_{i-1} - 1)/s_i)(\boldsymbol{\phi}_i - \boldsymbol{\phi}_{i-1})$.

while $i < I$ and $\|\boldsymbol{\phi}_i - \boldsymbol{\phi}_{i-1}\|_2 \geq \varepsilon$;

Set $\hat{\boldsymbol{\phi}}_\ell := \hat{\boldsymbol{\phi}}_\ell(\lambda_1, \lambda_2) = \boldsymbol{\phi}_i$.

4.2 Tuning Parameter Selection

To select the optimal MOD-VAR fit, we must choose (λ_1, λ_2) over a range of potential values. We describe three strategies for the selection of $\lambda_1, \lambda_2 > 0$: by-subject cross-validation, rolling-window cross-validation, and the adaptive moderator vector autoregression approach.

4.2.1 By Subject Cross-Validation

We introduce the by-subject cross-validation (BSCV) procedure for fitting our proposed MOD-VAR method when only shared and moderator-dependent components are modeled. Our J -fold BSCV procedure is as follows. First, we divide our set of subjects $[n]$ into J approximately equally sized partitions $\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_J$. Then, for each $j \in [J]$, and choice of tuning parameters $\lambda_1 > 0$, $\lambda_2 > 0$, we derive the solution to (6) based exclusively on the data of the subjects $[n] \setminus \mathcal{C}_j$, which we denote by $\hat{\boldsymbol{\Phi}}_{(BSCV,j)}(\lambda_1, \lambda_2) \in \mathbb{R}^{d \times (d+pd)}$. From this, we obtain the shared component $\hat{\boldsymbol{\Phi}}_{(BSCV,j)}^{(0)}(\lambda_1, \lambda_2) \in \mathbb{R}^{d \times d}$ and the moderator-dependent components $\{\hat{\boldsymbol{\Phi}}_{(BSCV,j)}^{(Mr)}(\lambda_1, \lambda_2)\}_{r=1}^p \subset \mathbb{R}^{d \times d}$. Then, for each subject $k \in \mathcal{C}_j$, we define the

cross-validation VAR coefficients for subject k based on the moderator vector $\mathbf{X}_k$ as

$$\hat{\Phi}_{(BSCV,j)}^{(\lambda_1, \lambda_2)}(\mathbf{X}_k) = \hat{\Phi}_{(BSCV,j)}^{(0)}(\lambda_1, \lambda_2) + \sum_{r=1}^p X_{kr} \hat{\Phi}_{(BSCV,j)}^{(Mr)}(\lambda_1, \lambda_2) \in \mathbb{R}^{d \times d}.$$

Here, $\hat{\Phi}_{(BSCV,j)}^{(\lambda_1, \lambda_2)}(\mathbf{X}_k)$ provides an estimation of the matrix of VAR coefficients corresponding to subject $k \in \mathcal{C}_j$ without using any data from subject k , instead relying only on the data from the subject set $[n] \setminus \mathcal{C}_j$. Finally, we define the by-subject cross-validation mean-squared forecasting error

$$\begin{aligned} BSCV.MSFE(\lambda_1, \lambda_2) &= \frac{1}{J} \sum_{j \in [J]} \frac{1}{|\mathcal{C}_j|} \sum_{k \in \mathcal{C}_j} \frac{1}{T_k - 1} \left\| \mathbf{Z}^{(k)} - \mathbf{Y}^{(k)} (\hat{\Phi}_{(BSCV,j)}^{(\lambda_1, \lambda_2)}(\mathbf{X}_k))' \right\|_F^2 \\ &= \frac{1}{J} \sum_{j \in [J]} \frac{1}{|\mathcal{C}_j|} \sum_{k \in \mathcal{C}_j} \frac{1}{T_k - 1} \sum_{t=2}^{T_k} \left\| \mathbf{Y}_t^{(k)} - \hat{\Phi}_{(BSCV,j)}^{(\lambda_1, \lambda_2)}(\mathbf{X}_k) \mathbf{Y}_{t-1}^{(k)} \right\|_2^2. \end{aligned}$$

We select the pair $(\lambda_1^*, \lambda_2^*)$ that minimizes the $BSCV.MSFE(\lambda_1, \lambda_2)$.

This approach has computational advantages compared to other procedures like rolling-window cross-validation, since it only requires J MOD-VAR fits for each candidate pair (λ_1, λ_2) . We emphasize that by-subject cross-validation can only be applied when no subject-specific components are incorporated in the model. If subject-specific structure is present in the model, data from subjects in $[n] \setminus \mathcal{C}_j$ cannot be used to estimate the GCN of subjects $k \in \mathcal{C}_j$. For cases in which some subject-specific edges may be present in the GCNs, we introduce the rolling-window cross-validation procedure.

4.2.2 Rolling-Window Cross-Validation

Rolling-window cross-validation is an alternative procedure for selecting the optimal MOD-VAR model, based on the method proposed in [22]. First, we select $T = \min_{k \in [n]} \{T_k - 1\}$, and we set $J = \lfloor T/3 \rfloor$. Then, for any $j \in [J]$, we use the observations $\cup_{k \in [n]} \{\mathbf{Y}_t^{(k)} : t \in [T_k - j - 2]\}$ as the training set. With this training data, we obtain the by-subject GCN estimators $\hat{\Phi}_{(RWCV,j)}^{(k; \lambda_1, \lambda_2)} \in \mathbb{R}^{d \times d}$ for each potential choice of $\lambda_1, \lambda_2 > 0$. Finally, we measure

the rolling-window cross-validation out-of-sample mean squared forecasting error as

$$RWCV.MSFE(\lambda_1, \lambda_2) = \frac{1}{J} \sum_{j \in [J]} \frac{1}{n} \sum_{k \in [n]} \|\mathbf{Y}_{T_k-j}^{(k)} - \hat{\Phi}_{(RWCV,j)}^{(k;\lambda_1,\lambda_2)} \cdot \mathbf{Y}_{T_k-j-1}^{(k)}\|_2^2.$$

We select the values $\lambda_1^*, \lambda_2^* > 0$ with the lowest RWCV.MSFE value.

4.2.3 Adaptive MOD-VAR

To improve upon the performance of our original estimator, we propose an adaptive MOD-VAR (adaMOD-VAR) estimator based on the adaptive LASSO [37]. For this, let $\hat{\Phi} = [\hat{\Phi}^{(0)}, \hat{\Phi}^{(M1)}, \dots, \hat{\Phi}^{(Mp)}, \hat{\Delta}^{(I1)}, \dots, \hat{\Delta}^{(In)}] \in \mathbb{R}^{d \times (d+pd+nd)}$ be an initial MOD-VAR estimator. Then, given a constant $\gamma > 0$, we define the adaptive weight for the shared and moderator-dependent coefficients $[\hat{\Phi}^{(0)}, \hat{\Phi}^{(M1)}, \dots, \hat{\Phi}^{(Mp)}]$ as

$$\omega_{ada}(\hat{\Phi})_{ij} = |\hat{\Phi}_{ij}|^{-\gamma}, \quad \forall 1 \leq i \leq d, 1 \leq j \leq d + pd. \quad (9)$$

For the subject-specific components $[\hat{\Delta}^{(I1)}, \dots, \hat{\Delta}^{(In)}]$, we select our adaptive penalty similar to the proposal of Fisher et al. [8]. First, based on $\hat{\Phi}$, we calculate the GCNs for each subject $\hat{\Phi}^{(1)}, \hat{\Phi}^{(2)}, \dots, \hat{\Phi}^{(n)} \in \mathbb{R}^{d \times d}$. Then, we calculate the entrywise median GCN $\tilde{\Phi} \in \mathbb{R}^{d \times d}$, where $\tilde{\Phi}_{ij}$ is the median of $\hat{\Phi}_{ij}^{(1)}, \hat{\Phi}_{ij}^{(2)}, \dots, \hat{\Phi}_{ij}^{(n)}$. Finally, we define the adaptive weight for $\Delta_{ij}^{(Ik)}$ as,

$$\omega_{ada}(\hat{\Phi})_{i,d(p+k)+j} = |\hat{\Delta}_{ij}^{(Ik)} - \tilde{\Phi}_{ij}|^{-\gamma}, \quad \forall 1 \leq i, j \leq d, 1 \leq k \leq n. \quad (10)$$

We then define the adaMOD-VAR estimator as the solution to the optimization problem

$$\hat{\Phi}_{ada}(\lambda) = \underset{\Phi \in \mathbb{R}^{d \times (d+pd+nd)}}{\operatorname{argmin}} \frac{1}{2N} \|\mathbf{Z} - \mathbf{W}\Phi'\|_F^2 + \lambda \sum_{i=1}^d \sum_{j=1}^{d+pd+nd} \omega_{ada}(\hat{\Phi})_{ij} \cdot |\Phi_{ij}|, \quad \text{for } \lambda > 0. \quad (11)$$

While the adaptive MOD-VAR estimator introduced in (11) incorporates shared, moderator-dependent and subject-specific components, the adaptive MOD-VAR procedure can also be defined for models that include only moderator-dependent or subject-specific components by selecting the blocks of $\omega_{ada}(\hat{\Phi})$ accordingly.

5 Theoretical Properties

We explore the theoretical properties of the proposed MOD-VAR estimator and establish high-dimensional consistency guarantees. In particular, we are interested in the estimation errors under the Frobenius and L_1 norms $\|\hat{\Phi} - \Phi^*\|_F$ and $\|\hat{\Phi} - \Phi^*\|_1$. To establish these results, we assume the true parameter matrix Φ^* is s -sparse: $\|\Phi^*\|_0 := |\{(i, j) : \Phi_{ij}^* \neq 0\}| \leq s$. To simplify our exposition, we assume that all time series have the same number of time observations, *i.e.*, $T := T_1 = T_2 = \dots = T_n$. We denote the total number of observations by $N := n(T - 1)$.

To establish our guarantees, consider the following set of assumptions.

Assumption 1 (Bounded moderators). *Assume the moderator vectors $\mathbf{X}, \mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$ are independent, identically distributed and entrywise bounded random vectors satisfying $|X_i| \leq B$ almost surely, with mean zero, and covariance matrix $\Sigma_X = \mathbb{E}[\mathbf{X}\mathbf{X}']$. Furthermore, assume that for some constant $0 < \eta < 1$, $\sup_{\|\mathbf{x}\|_\infty \leq B} \|\Phi^*(\mathbf{x})\|_2 < \eta < 1$.*

Assumption 2 (Curvature condition). *Let $\Lambda_Y := \inf_{\|\mathbf{x}\|_\infty \leq 1} \Lambda_{\min}(\text{Cov}(\mathbf{Y}_t^k | \mathbf{X}_k))$, and $\alpha = \Lambda_Y \cdot \min\{1, \Lambda_{\min}(\Sigma_X)\}/2$. There for a suitable choice of constants $C > 0$ and $t > 0$,*

$$\alpha > Cs \left(\sqrt{\frac{\eta(\log(pd^2) + t)}{n}} + \frac{K \log(2n)(\log(pd^2) + t)}{n} \right).$$

Our Assumption 1 requires that the moderator variables are entrywise bounded. Prior works have imposed similar bounded moderator assumptions [34]. In applications, moderators often correspond to categorical or bounded score variables. For example, in the analysis of multi-subject fMRI data, moderators of interest are often variables such as biological sex and age, or prior health conditions, which can all be coded and normalized as discrete, centered, and bounded variables. The condition $\sup_{\|\mathbf{x}\|_\infty \leq B} \|\Phi^*(\mathbf{x})\|_2 < \eta < 1$ serves to ensure that the VAR process is stable for every admissible moderator vector [2].

The convergence of our MOD-VAR estimator relies on ensuring sufficient curvature of the matrix $\mathbf{W}'\mathbf{W}/N \in \mathbb{R}^{(d+pd) \times (d+pd)}$. Since the design matrix $\mathbf{W}$ is composed of the blocks $(\mathbf{X}_k^*)' \otimes \mathbf{Y}^{(k)}$, where $\mathbf{X}_k^* = (1, \mathbf{X}_k')' \in \mathbb{R}^{p+1}$, the minimum eigenvalue of $\mathbb{E}[\mathbf{W}'\mathbf{W}/N]$ can be controlled by imposing bounds on the minimum eigenvalue $\Lambda_{\min}(\Sigma_{X^*}) = \min\{1, \Lambda_{\min}(\Sigma_X)\}$ and the conditional covariances $\text{Cov}(\mathbf{Y}_t^{(k)} | \mathbf{X}_k)$. Similar minimum eigenvalue controls are commonly used in other high-dimensional consistency results [19; 2; 31].

Under Assumptions 1 and 2, we derive high-dimensional guarantees of convergence.

Theorem 1. *Consider multi-subject time-series data $\{(\mathcal{Y}_k, \mathbf{X}_k)\}_{k=1}^n$ that follow the MOD-VAR model (2). Furthermore, suppose Assumption 1 holds and $n > 16 \log(2n) \log(pd^2)$. Let $C_M := \max_{k \in [n]} \{\Lambda_{\max}(\Sigma_{\varepsilon}^{(k)})\} (2 + (1 + 4\eta)/(1 - \eta)^2)$. Then, there exist a constant $C > 0$ depending on B and C_M such that Assumption 2 is satisfied for a given $t > 0$. Furthermore, if $\lambda := \lambda_1 = \lambda_2$ satisfies the inequality $\lambda > 16 \max\{1, B\} \pi C_M \sqrt{\log(pd^2)/N}$. Then, with probability at least $1 - 3e^{-t} - (6/(pd^2))^2$,*

$$\|\hat{\Phi} - \Phi^*\|_F \leq \frac{3\sqrt{s}\lambda}{\alpha}; \quad (12)$$

$$\|\hat{\Phi} - \Phi^*\|_1 \leq \frac{12s\lambda}{\alpha}. \quad (13)$$

Our Theorem 1 provides guarantees on the convergence of our proposed MOD-VAR, even in high-dimensional scenarios. First, note that our bound on $\|\hat{\Phi} - \Phi^*\|_1$ immediately implies a bound on the VAR coefficient matrix estimation of each subject via,

$$\begin{aligned} \max_{k=1, \dots, n} \|\hat{\Phi}(\mathbf{X}_k) - \Phi^*(\mathbf{X}_k)\|_1 &\leq \|\hat{\Phi}^{(0)} - \Phi^{(0)*}\|_1 + \max_{k=1, \dots, n} \sum_{i=1}^p |X_{ki}| \|\hat{\Phi}^{(Mi)} - \Phi^{(Mi)*}\|_1 \\ &\leq \max\{1, B\} \cdot \|\hat{\Phi} - \Phi^*\|_1. \end{aligned}$$

Therefore, the maximum subject-specific L_1 -norm error for our proposed estimator converges to zero whenever $\log(pd^2) = o(N)$. Notice that the rate $O(Bs\sqrt{(\log(p) + 2\log(d))/N})$ described in (13) depends on the total sample size N , instead of the subject-level sam-

ple size T . In particular, if $B = O(1)$ and $\beta := \beta_n > 0$ satisfies $(\beta + 1)/n = o(1)$ and $p = O(d^\beta)$, we have $\max_{k=1,\dots,n} \|\hat{\Phi}(\mathbf{X}_k) - \Phi^*(\mathbf{X}_k)\|_1 = O(s\sqrt{(\beta + 1)/n} \cdot \sqrt{\log(d)/T})$. This improves upon the estimation rate $O(s\sqrt{\log(d)/T})$ of individual VAR estimators [2] when $n \rightarrow \infty$. For example, for any $0 \leq \alpha < 1$, and $p = O(d^{n^\alpha})$, we have $\max_{k=1,\dots,n} \|\hat{\Phi}(\mathbf{X}_k) - \Phi^*(\mathbf{X}_k)\|_1 = O\left(sn^{(\alpha-1)/2} \cdot \sqrt{\log(d)/T}\right) = o(s\sqrt{\log(d)/T})$, demonstrating improved performance for both high-dimensional d and number of moderators p .

The convergence rate of the MOD-VAR method scales with the total number of observations N . Thus, consistency is attained even when the number of observations per subject T is limited, as long as the number of subjects n is large enough. To establish convergence, the number of subjects n must satisfy $\log(pd^2) = o(n/\log(n))$, which is connected to the sufficient curvature condition of Assumption 2. This condition on n ensures sufficient heterogeneity in the moderator data, which facilitates the consistent estimation of Σ_X and the GCN moderator-dependent effects.

6 Numerical Simulations

To assess the performance of our proposed MOD-VAR estimator, and compare it to methods in the literature, we perform a set of numerical simulation experiments.

6.1 Data Generating Method

To generate our data, we select a dimension $d \in \{10, 20\}$, number of available moderator variables $p \in \{2, 5\}$, number of subjects $n \in \{15, 20, 25\}$, and number of available observations per time series $T \in \{30, 60, 90\}$.

To simulate data, we first generate a matrix of moderator variables $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n]' \in \mathbb{R}^{n \times p}$, as well as VAR coefficient matrices of shared, moderator-dependent and subject-

specific components $\Phi^{(0)*} \in \mathbb{R}^{d \times d}$, $\{\Phi^{(Mr)*}\}_{r=1}^p \subset \mathbb{R}^{d \times d}$, and $\{\Delta^{(Ik)*}\}_{k=1}^n \subset \mathbb{R}^{d \times d}$, respectively. Once these matrices are generated, for each $k \in [n]$, we define the matrix of VAR coefficients for subject k with moderators $\mathbf{X}_k = (X_{k1}, \dots, X_{kp})'$ as,

$$\Phi^{(k)*} = \Phi^{(0)*} + \sum_{r=1}^p X_{kr} \Phi^{(Mr)*} + \Delta^{(Ik)*} \in \mathbb{R}^{d \times d}.$$

For each $k \in [n]$, we generate the time-series data $\{\mathbf{Y}_t^{(k)}\}_{t=1}^{T+403}$, by setting $\mathbf{Y}_0^{(k)} = \mathbf{0}$, and $\mathbf{Y}_t^{(k)} = \Phi^{(k)*} \mathbf{Y}_{t-1}^{(k)} + \boldsymbol{\varepsilon}_t^{(k)}$, where $\{\boldsymbol{\varepsilon}_t^{(k)}\}_{t=1}^{T+403} \stackrel{i.i.d.}{\sim} N_d(\mathbf{0}, (0.1) \cdot \mathbf{I}_d)$. Then, we discard the first 400 time points of each time series. The additional 3 time measurements are used to evaluate out-of-sample forecasting performance.

Our data generating process requires generating the moderator matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$, and the matrices $\Phi^{(0)*}$, $\{\Phi^{(Mr)*}\}_{r=1}^p$ and $\{\Delta^{(Ik)*}\}_{k=1}^n$. To generate $\mathbf{X} \in \mathbb{R}^{n \times p}$ we set $\mathbf{X}_k = (X_{k1}, \dots, X_{kp}) \stackrel{i.i.d.}{\sim} (0.25) \cdot \delta_0 + (0.75) \cdot \text{Unif}([-1, -0.5] \cup [0.5, 1])$, where δ_x denotes the atom probability measure at $x \in \mathbb{R}$. To generate the shared, moderator-dependent and subject-specific components, we introduce $q_0, q_m, q_\Delta \in [0, 1]$ the shared, moderator-dependent and subject-specific connection probabilities, respectively. With these parameters in place, we generate our data as follows. First, we generate the shared component $\Phi^{(0)*} \in \mathbb{R}^{d \times d}$, moderator dependent components $\{\Phi^{(Mr)*}\}_{r=1}^p \subset \mathbb{R}^{d \times d}$, and subject-specific components $\{\Delta^{(Ik)*}\}_{k=1}^n \subset \mathbb{R}^{d \times d}$ as follows:

$$\begin{aligned} \{\Phi_{ij}^{(0)*}\}_{i,j=1}^d &\stackrel{i.i.d.}{\sim} (1 - q_0)\delta_0 + q_0 \text{Unif}([-0.9, -0.5] \cup [0.5, 0.9]); \\ \{\Phi_{ij}^{(Mr)*}\}_{i,j=1}^d &\stackrel{i.i.d.}{\sim} (1 - q_m)\delta_0 + q_m \text{Unif}([-0.9, -0.5] \cup [0.5, 0.9]); \\ \{\Delta_{ij}^{(Ik)*}\}_{i,j=1}^d &\stackrel{i.i.d.}{\sim} (1 - q_\Delta)\delta_0 + q_\Delta \text{Unif}([-0.9, -0.5] \cup [0.5, 0.9]). \end{aligned}$$

Note that the parameters q_0 , q_m and q_Δ control the level of sparsity of the matrices $\Phi^{(0)*}$, $\{\Phi^{(Mr)*}\}_{r=1}^p$ and $\{\Delta^{(Ik)*}\}_{k=1}^n$, respectively. In particular, notice that if $q_m = 0$, $\Phi^{(Mr)*} = \mathbf{0}$ for all $r \in [p]$. From this, when $q_m = 0$ moderator variables do not have any effect on the connectivity pattern across the GCNs. Similarly, $q_\Delta = 0$ implies that $\Delta^{(Ik)*} = \mathbf{0}$ for

all $k \in [n]$, which corresponds to an absence of subject-specific components in the GCNs. In our simulations, we select different choices of q_0 , q_m and q_Δ to explore the impact that moderator-dependent and subject-specific components have on estimation accuracy.

6.2 Methods and Performance Metrics

We compare our proposed MOD-VAR to methods in the literature of multiple time-series modeling. First, we consider the vector autoregressive (VAR) model [2] applied separately to each time series. This method is implemented via the `bigVAR` R package [21]. We also compare to the original and adaptive Multi-VAR methods [7], implemented via the `multivar` package. Finally, we compare our MOD-VAR and adaMOD-VAR methods, fitted with cross-validation.

To evaluate the performance, we consider the mean L_1 and Frobenius estimation errors, and out-of-sample 1-step root mean squared forecasting errors, given by

$$\begin{aligned} E_{L_1}(\hat{\Phi}, \Phi^*) &= \frac{1}{n} \sum_{k=1}^n \|\hat{\Phi}^{(k)} - \Phi^{(k)*}\|_1, \\ E_F(\hat{\Phi}, \Phi^*) &= \frac{1}{n} \sum_{k=1}^n \left\| \hat{\Phi}^{(k)} - \Phi^{(k)*} \right\|_F, \\ RMSFE(\hat{\Phi}) &= \frac{1}{n} \sum_{k=1}^n \left(\frac{1}{3} \sum_{h=1}^3 \|\mathbf{Y}_{T+h}^{(k)} - \hat{\Phi}^{(k)} \mathbf{Y}_{T+h-1}^{(k)}\|_2^2 \right)^{1/2}, \end{aligned}$$

respectively. We report the average performance over 100 simulation replicates. We show additional results comparing the mean true positive rate and mean false positive rate of GCN recovery, and computational time in our Supplementary Materials.

6.3 Results: Only Moderator-Driven Heterogeneity

First, we explore numerical experiments in which all the GCN heterogeneity across subjects consists of moderator-dependent components. This corresponds to the condition $\Delta^{(Ik)*} = 0$

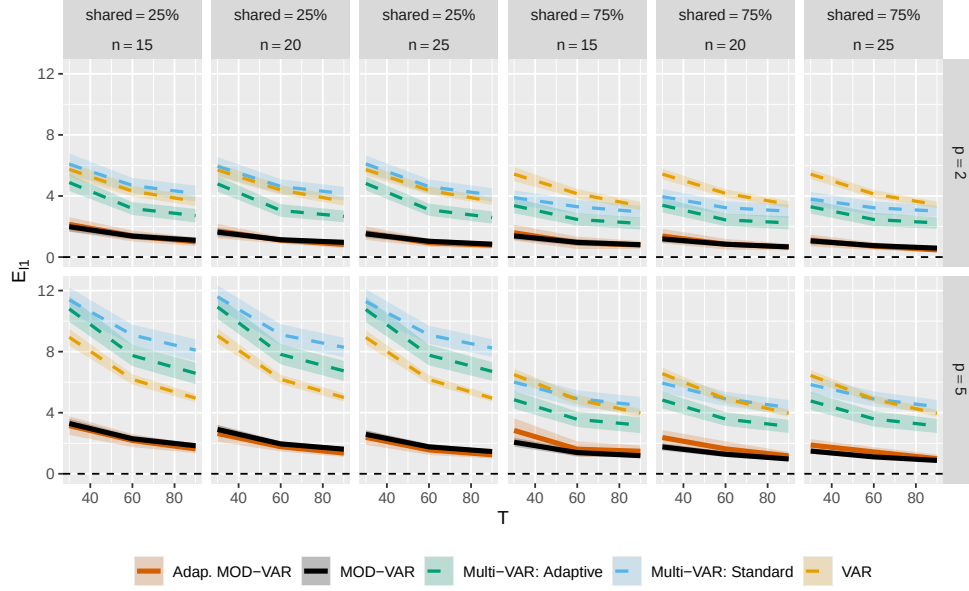


Figure 3: Visualization of the mean L_1 -norm parameter estimation error for our proposed MOD-VAR and methods in the literature for data where the heterogeneity across subjects is fully explained by moderator variables. We explore performance for $d = 10$, varying n , T , p and the proportion of shared/moderated components. Our proposed MOD-VAR and adaptive MOD-VAR attain substantially lower error for all choices of number of subjects n , moderators p and percentage of shared structure in the GCNs.

for all $k \in [n]$, or $q_\Delta = 0$. We explore the values $(q_0, q_m) \in \{(0.075, 0.025), (0.025, 0.075)\}$.

In Figures 3 and 4 we visualize the average E_{L_1} and $RMSFE$ for $d = 10$.

As we observe in Figure 3, both the MOD-VAR and adaptive MOD-VAR attain the best performance in terms of L_1 -norm error across all simulation scenarios. Depending on the degree of shared structure of the GCNs, the second lowest L_1 -norm error is attained by the individual VAR or the Multi-VAR methods. For example, when 75% of the GCN is shared, *i.e.* $q_0 = 0.075$, or $p = 2$, the adaptive Multi-VAR is the second best performing method. On the other hand, in our scenario with the highest heterogeneity, *i.e.* only 25% of entries are shared and $p = 5$, the VAR method achieves second best performance. Our proposed MOD-VAR attains a substantially better performance, since it can model joint structure, and uses moderator data to drastically reduce the parameter complexity of estimating the heterogeneous structure. In a similar manner, we observe our MOD-VAR attains the best

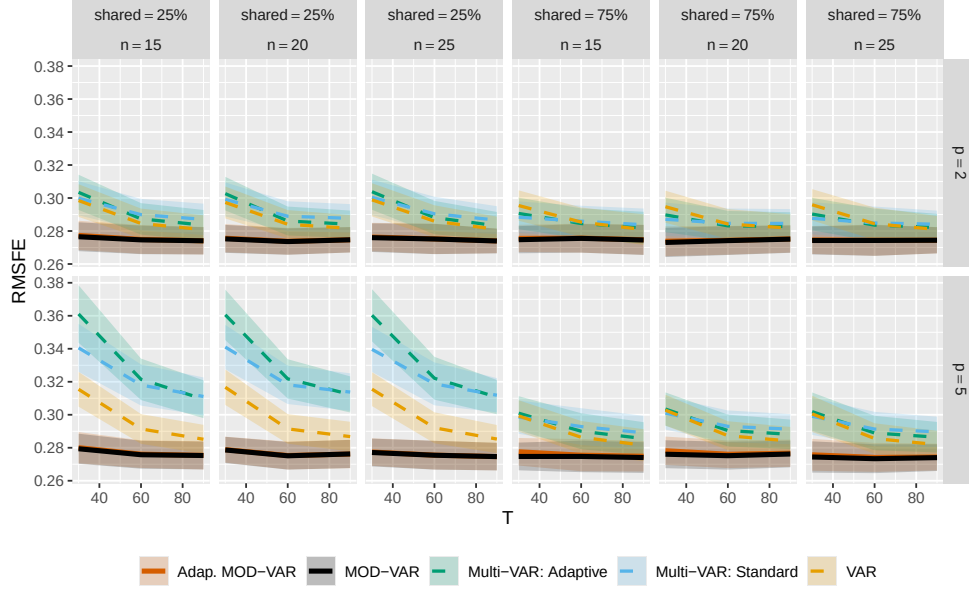


Figure 4: Visualization of the out-of-sample 1-step RMSFE for our proposed MOD-VAR and methods in the literature for data where the heterogeneity across subjects is fully explained by moderator variables. We explore performance for $d = 10$, varying n , T , p and the proportion of shared/moderated components. Our proposed MOD-VAR and adaptive MOD-VAR attain substantially lower error for all choices of number of subjects n , moderators p and percentage of shared structure in the GCNs.

performance in terms of out-of-sample 1-step RMSFE, as shown in Figure 4.

6.4 Results: Moderator and Subject-Specific Heterogeneity

To further explore the performance of our estimation procedure when both moderator-dependent and subject-specific components are present in the GCNs, we present additional numerical experiments. For this, we set $(q_0, q_m) \in \{(0.075, 0.025), (0.025, 0.075)\}$ and $q_\Delta = 0.03$, which corresponds to a positive probability of non-zero subject-specific components.

We visualize the L_1 -norm estimation error and RMSFE for $d = 10$ in Figures 5 and 6, respectively. As observed in both Figures 5 and 6, our proposed MOD-VAR attains the best performance in terms of both parameter estimation and forecasting error, for all number of available time points T , number of subjects n and moderators p . These results confirm the desirable performance of our proposed MOD-VAR in scenarios in which both

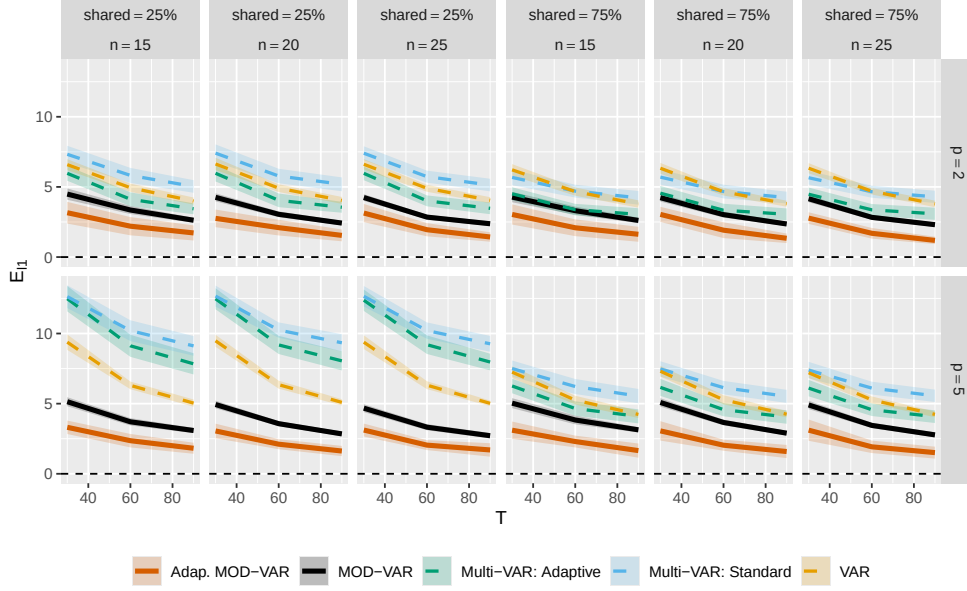


Figure 5: Visualization of the mean L_1 -norm parameter estimation error for our proposed MOD-VAR and methods in the literature for data where the subject heterogeneity is a combination of moderator and subject-specific components. We explore performance for $d = 10$, varying n , T , p and the proportion of shared/moderated components. Our proposed adaptive MOD-VAR attains substantially lower error for all choices of number of subjects n , moderators p and percentage of shared structure in the GCN.

moderator-dependent and subject-specific components are present in heterogeneous multi-subject data. We provide additional simulation results in our Supplementary Materials.

7 Analysis of fMRI Time-Series Data

We analyze a resting-state fMRI time-series dataset measured over multiple participants, reported in Hwang et al. [12]. Critically, this dataset contains intensive resting state fMRI measurements for $n = 26$ participants, with $T = 150$ observations per subject. The dataset consists of measurements of brain activity for regions composing the Frontoparietal Network (dorsolateral prefrontal cortex, posterior parietal cortex, dorsal anterior cingulate cortex/pre-supplementary motor area), default mode network (medial prefrontal cortex, posterior cingulate cortex/precuneus, inferior parietal lobule, medial temporal lobe), and

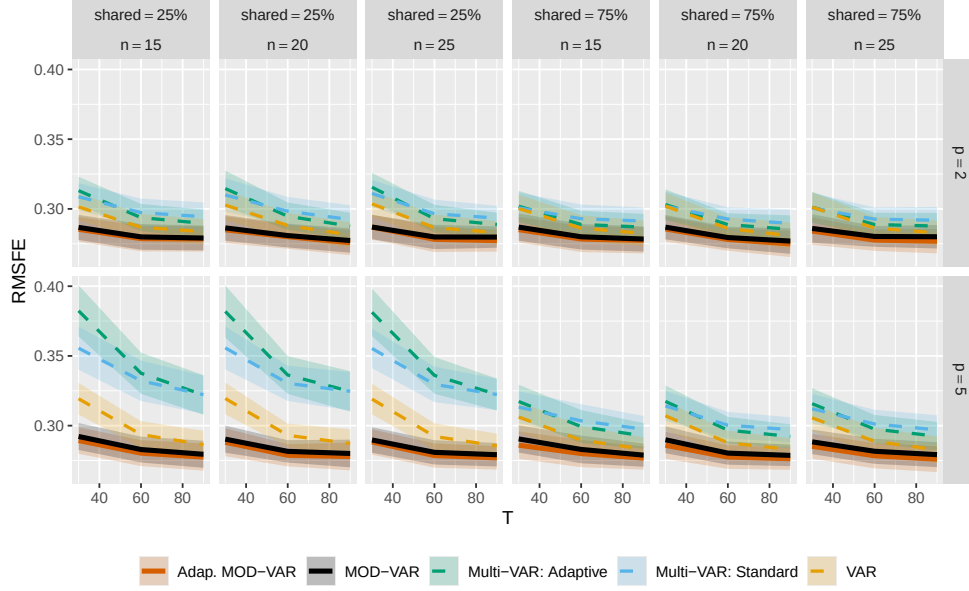


Figure 6: Visualization of the RMSFE for our proposed MOD-VAR, and methods in the literature for data where the subject heterogeneity is a combination of moderator effects and subject-specific connections. We explore performance for $d = 10$, varying n , T , p and the proportion of shared/moderated components. Our proposed adaptive MOD-VAR attains substantially lower error for all choices of number of subjects n , moderators p and percentage of shared structure in the GCN.

salience network (anterior insula, anterior cingulate cortex). This gives a total of $d = 9$ variables. In addition, this dataset contains $p = 4$ moderators, including participant’s age, biological sex, locus coeruleus volume (LC-volume), and locus coeruleus contrast-to-noise ratio (LC-CNR). It has been demonstrated that both age [1; 10] and locus coeruleus structure [35; 28] are associated with activity within these different functional brain networks.

We compare the out-of-sample forecasting accuracy between VAR, Multi-VAR, and MOD-VAR models on this fMRI dataset through the following experiment. For each method, we train each model on the data corresponding to the first T_{train} observations per subject, where $10 \leq T_{train} \leq 100$, resulting in the estimated by-subject autoregressive coefficients $\hat{\Phi}^{[n]} = \{\hat{\Phi}^{(k)}\}_{k \in [n]}$. Then, we evaluate the out-of-sample J -step RMSFE, given by

$$RMSFE_J(\hat{\Phi}^{[n]}) := \frac{1}{n} \sum_{k=1}^n \left(\frac{1}{50} \sum_{h=1}^{50} \|\mathbf{Y}_{T_{train}+h}^{(k)} - (\hat{\Phi}^{(k)})^J \mathbf{Y}_{T_{train}+h-J}^{(k)}\|_2^2 \right)^{1/2}.$$

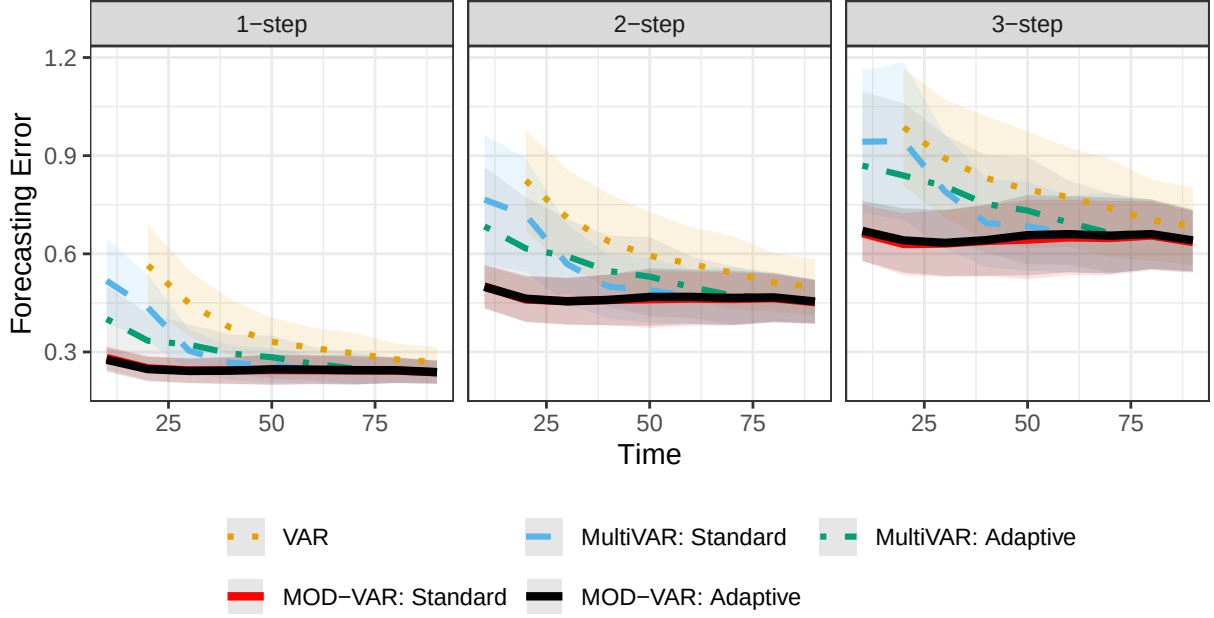


Figure 7: Comparison of the out-of-sample RMSFE of our proposed MOD-VAR, as well as the Multi-VAR and individual VAR models on the resting state fMRI dataset. By incorporating moderator information, our proposed MOD-VAR attains the lowest error for 1-, 2-, and 3-step forward forecasting performance.

To assess the quality of forecasting of the models, we consider the 1-, 2- and 3-step out-of-sample RMSFE performance.

We visualize the numerical RMSFE across methods in Figure 7. Both the standard and adaptive MOD-VAR models exhibited the lowest out-of-sample RMSFE across 1-, 2-, and 3-step predictions, thereby outperforming the standard VAR and Multi-VAR models. This result suggests that incorporating moderator-dependent components in the model substantially aids forecasting performance. We observe that both the MOD-VAR and Multi-VAR obtain a better performance than individually estimating VAR models, which strongly suggests the presence of shared structure across participants. Despite the advantage enjoyed by both the Multi-VAR and MOD-VAR methods, Multi-VAR must model a total of $(n + 1)d^2 = 2187$ parameters. On the other hand, with $p = 4$ moderators, our MOD-VAR estimator using only shared and moderator-dependent components must esti-

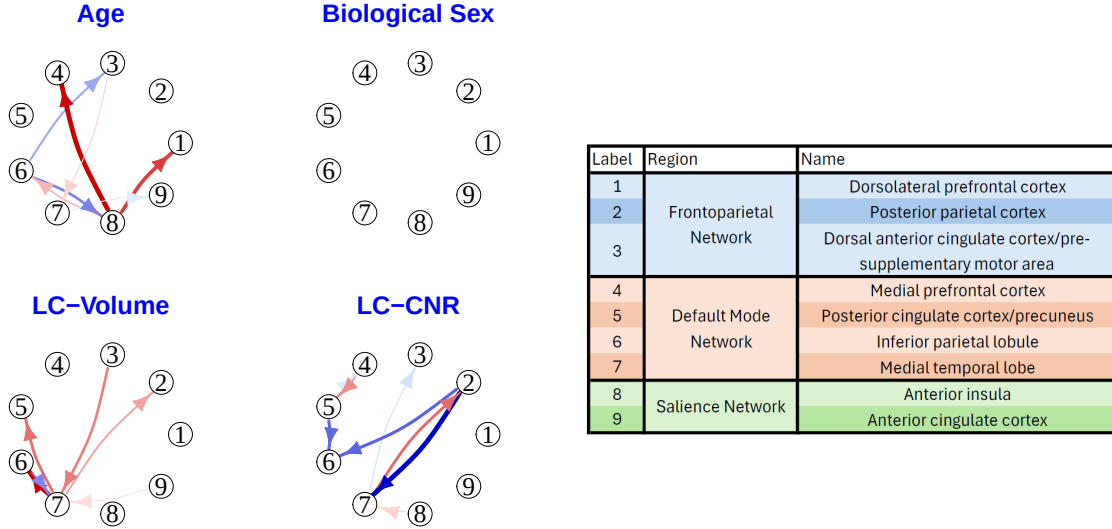


Figure 8: Visualization of the moderator effects estimated by our MOD-VAR method. Age moderates interactions of the anterior insula in the salience network. Furthermore, locus coeruleus volume has network moderation effects, especially for the default mode network’s medial temporal lobe. Locus coeruleus CNR moderates the posterior parietal cortex in the frontoparietal network. We found no moderation effects for biological sex on the network dynamics.

mate $(p + 1)d^2 = 405$ parameters. Thus, by jointly modeling the multi-subject data, and reducing complexity by using moderator data, we attain improved forecasting performance.

Next we examine the estimated moderator components obtained by our MOD-VAR method in Figure 8. Increasing age was associated with reduced predictive signaling between regions in the salience network (particularly anterior insula) and both the frontoparietal network (dorsolateral prefrontal cortex) and the default mode network (medial prefrontal cortex). Further, predictive strength increased between the inferior parietal lobule and the anterior insula.

We observed no moderation effects of biological sex on the predictive network dynamic.

Higher LC-volume was associated with reduced predictive signaling within the default mode network regions, specifically from the medial temporal lobe to both the posterior cingulate cortex/precuneus, inferior parietal lobule. Reduced predictive strength was also ob-

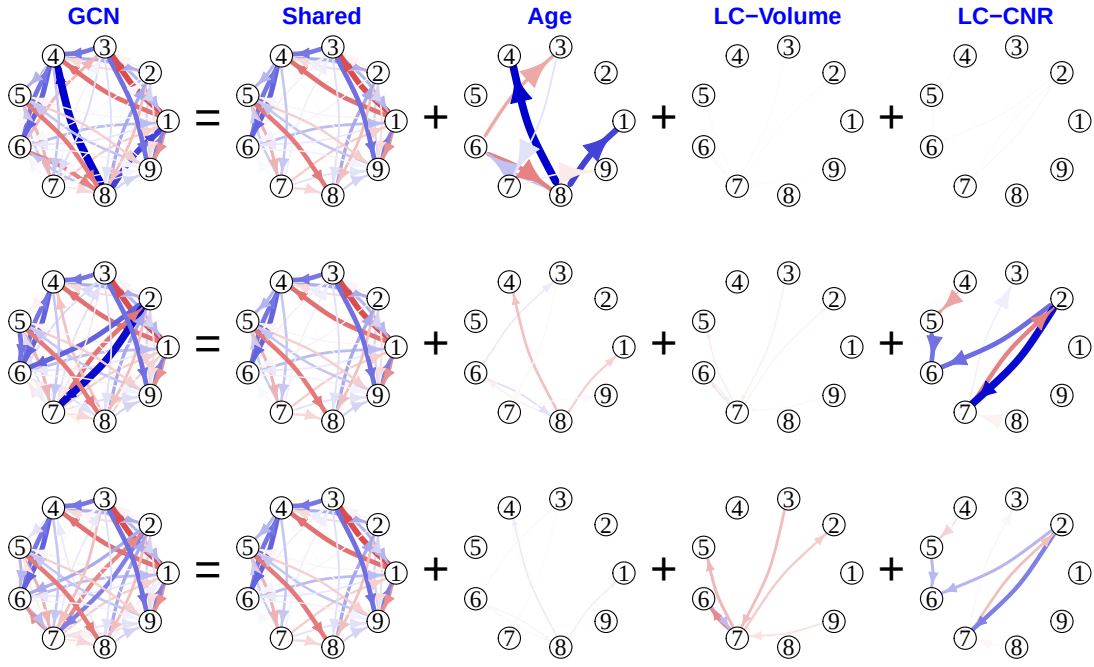


Figure 9: Visualization of the GCNs and shared/moderated GCN components for three subjects in our data. The GCNs across subjects exhibit heterogeneity driven by age, as well as locus coeruleus volume and contrast-to-noise ratio.

served from the frontoparietal network (dorsal anterior cingulate cortex/pre-supplementary motor area) to the default mode network (medial temporal lobe). Further, there was an increase in predictive strength from the inferior parietal lobule to the medial temporal lobe.

Higher LC-CNR was associated with increased predictive signaling from the frontoparietal network (posterior parietal cortex) to the default mode network (both the medial temporal lobe and the inferior parietal lobule). Increased predictive strength was also found from the posterior cingulate cortex/precuneus to the inferior parietal lobule. Conversely, elevated LC-CNR was associated with reduced predictive signaling from the medial prefrontal cortex to the posterior cingulate cortex/precuneus, as well as from the medial temporal lobe to the posterior parietal cortex.

We visualize the heterogeneity of our estimated GCNs via the decomposition into shared and moderator-dependent components in Figure 9. We observe that the GCNs exhibit major differences across subjects. Furthermore, the components corresponding to age, LC-

volume and LC-CNR vary in magnitude, depending on the moderator values.

While the biological interpretations of these effects are outside the scope of the current paper, the results are somewhat consistent with theories of age-related declines in the locus coeruleus (LC) and its associated role in moderating functional brain network activity. Furthermore, no biological sex effects were found. On the one hand, prior work examining biological sex and resting state networks has found relationships while other work has suggested that these findings are highly dependent on data processing decisions. Finally, LC metrics were associated with predictive signaling in areas of the frontoparietal network and default mode network in a manner consistent with prior literature [24].

8 Discussion

In this work, we introduce the moderator vector autoregressive model (MOD-VAR) for estimating the GCNs associated with multi-subject multivariate time-series data. Our proposed MOD-VAR can identify common components across heterogeneous GCNs, and quantify whether additional components may be explained by external moderator variables. Our procedure outperforms methods in the literature in terms of both GCN estimation and forecasting error, as demonstrated by our extensive numerical experiments. Furthermore, our theoretical guarantees show that MOD-VAR attains improved non-asymptotic rates of convergence compared to previous results in the literature [2; 7].

This work opens a variety of new avenues of research. In particular, we consider it important to further explore applications of our proposed MOD-VAR to other multi-subject studies in psychology and neuroscience, similar to the works of [14; 25; 5]. Furthermore, our proposed method opens a variety of new avenues of methodological research. For example, it would be of interest to test the statistical significance of the effect of moderators on

GCNs and explore MOD-VAR extensions to higher-order VAR models, heavy-tailed errors, and moderator effects beyond the additive assumption.

Supplementary Materials

We provide proofs, additional discussions and simulations in the Supplementary Materials.

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